

Agentic Delegation

and the Language Frontier

Quispe & Xu (2026), arXiv:2605.25438v2

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1. Citation check: the discrepancy is real

Current v2: Alexander Quispe and Kevin Xu, “Agentic Delegation and the Language Frontier of Software Developers: A Model and Evidence from Claude Code on GitHub,” July 8, 2026.

Earlier circulation

“Coding Beyond Your Training: Claude Code and the Technological Frontier of Software Developers,” Alexander Quispe only.

Correct status

Preliminary, unrefereed **arXiv economics preprint**; not NBER. v1: May 25. v2: July 7; PDF dated July 8.

2. Question, claim, and non-claim

Research question

Does agentic AI expand the languages in which a developer can ship working code, beyond accelerating work in familiar languages?

Claim: the developer's observed **production frontier** expands.

Non-claim: the developer learns to code unassisted in the new language. Delegated output remains joint production: the human selects, specifies, verifies, and ships.

Unit: developer $i \times$ language $k \times$ month t .

3. Three capabilities

Solo S

Expected human match $s\mu$, activation cost b , and CARA risk penalty from match uncertainty.

Augmentation C

Adds $\gamma s - r_C$: suggestions are useful only with language-specific foothold s .

Delegation D

Agent executes share λ ; output uses general ability a , capability $z(A)$, verification cost κ , and agent-error risk.

One modelling choice drives the result: **AI supplies delegated execution, not merely a substitutable productivity input.**

4. The developer's choice is not effort

$$M^1 = \{S, C\}, \quad M^2 = \{S, C, D\},$$
$$V_{ikt}^g = \max_{m \in M^g} V_{ikt}^m, \quad Z_{ikt}^g = 1\{V_{ikt}^g \geq 0\}, \quad N_{it}^g = \sum_k Z_{ikt}^g.$$

Economic decision

Given an opportunity, choose the mode with the highest certainty-equivalent surplus, then enter only if that surplus is nonnegative.

Key point: there is no endogenous continuous effort variable. An “agent effort problem” belongs to a different paper/model.

5. Thresholds and the delegation advantage

$$T^S = b - s\mu + \frac{\rho s^2}{2\pi}, \quad T^1 = T^S - \max\{0, \gamma s - r_C\}.$$

For unfamiliar k , Assumption 1 imposes $\gamma s - r_C \leq 0$, so $T^1 = T^S$. Define

$$B \equiv T^S - T^D = \underbrace{\lambda[az(A) - s\mu]}_{\text{execution substitution}} - \underbrace{\kappa(a, s) - r_D}_{\text{verification + compute}} + \underbrace{\frac{\rho}{2} \left[\frac{(2\lambda - \lambda^2)s^2}{\pi} - \sigma_D^2 \right]}_{\text{risk substitution}}.$$

Condition

Delegation changes entry only when $B > 0$. Better verification technology can enlarge the band; it does not guarantee a positive band everywhere.

6. Propositions 1–2

Proposition 1 (weak frontier expansion).

$$M^1 \subset M^2 \implies Z_{ikt}^2 \geq Z_{ikt}^1 \implies N_{it}^2 \geq N_{it}^1.$$

The agent can always be ignored; strict expansion is not automatic.

Proposition 2 (activation band). For unfamiliar k , Assumption 1 and $B > 0$ imply

$$Z_{ikt}^2 - Z_{ikt}^1 = 1\{T_{ikt}^D \leq \omega_{ikt} < T_{ikt}^1\}.$$

With continuous conditional CDF F , activation probability is $F(T^1) - F(T^D)$.

7. Proposition 3: stock and flow

If constant first-use hazards satisfy $p_{ik}^2 \geq p_{ik}^1$, then

$$\Delta C_i(h) = \sum_{k \in U_i} [(1 - p_{ik}^1)^{h+1} - (1 - p_{ik}^2)^{h+1}] \geq 0.$$

Closed frontier

If $p^1 = 0 < p^2$, the cumulative gap grows strictly and concavely over the observed horizon.

Endpoint/domain caveat

With $p^1 > 0$, monotonic growth needs the paper's additional “no catch-up” inequality; $p^2 \geq p^1$ alone gives nonnegativity, not monotonicity.

8. Data construction

Object	Size	
Window	Jan 2024–Apr 2026 (28 months)	
Initial / estimation developers	5,825 / 5,346	
Treated / not-yet-treated control	2,813 / 2,533	Treatment: first
Developer-month rows	149,688	
Commits / developer–repository pairs	3.2m / 133k	
Changed files classified by Linguist	57m	
Claude-coauthored detection universe	7.8m commits	

machine-readable Claude co-author trailer. Outcomes use non-merge commits and programming-language files.

9. Empirical strategy

Estimator

Callaway–Sant’Anna group-time ATT, doubly robust, not-yet-treated comparisons, one-month anticipation, varying base periods, 1,000 developer-clustered multiplier-bootstrap draws.

Four frontier outcomes: monthly distinct languages, newly used languages, cumulative languages, and Shannon entropy. Repositories and commits are diagnostics.

Interpretation discipline

Conditional parallel trends handles staggered timing and observed pre-period differences. It does not handle a new-project shock that both induces Claude adoption and changes languages.

10. Main numerical results

Outcome	$e = 0$	$e = 1$	$e = 2$	Simple ATT	Active languages
Active languages	2.528 (.063)	1.227 (.064)	.693 (.067)	1.178 (.052)	
Newly used	1.193 (.051)	.126 (.034)	-.018 (.038)	.294 (.030)	
Entropy	.382 (.009)	.189 (.011)	.102 (.012)	.172 (.010)	
Cumulative	1.604 (.054)	1.892 (.063)	2.072 (.086)	1.930 (.068)	
Repositories	1.494 (.058)	.838 (.057)	.449 (.057)	.742 (.057)	
Monthly commits	35.080 (2.085)	21.046 (2.292)	12.110 (2.609)	18.677 (2.269)	

roughly triple relative to the .90 pre-mean; new languages are nearly four times the .31 pre-flow.
Cumulative languages have significant pre-trends and are descriptive.

11. Mechanism evidence: headroom

Adoption-month newly used languages	Specialist	Generalist
High pre-adoption activity	.981 (.084)	.301 (.100)
Low pre-adoption activity	2.388 (.099)	1.015 (.092)

Matches Proposition 4's headroom margin

Within activity halves, narrow-portfolio specialists have more unfamiliar candidates and show much larger first-use effects.

The general-ability prediction is not cleanly supported: commit volume is a poor ability proxy and low-volume users show larger measured effects.

12. Robustness and what it cannot rule out

Survives:

- ▶ removing first-Claude language;
- ▶ excluding all Claude commits;
- ▶ activity conditioning;
- ▶ stricter pre-activity filters;
- ▶ competing-agent screens;
- ▶ placebo dates.

Still unresolved

A developer may start an unfamiliar-language project and adopt Claude because of that same project. The observed jump can combine treatment with selection on a time-varying shock.

Correct label: sharp, robust event-time association consistent with delegation.

13. Symbolic check and handwritten derivation

SymPy result

Exact identity

```
simplify((T_s - T_d) - B) == 0
```

Rational example: $T^D = -487/1250$,
 $T^S = 81/100$, and band width
 $B = 2999/2500 > 0$.

The image shows a handwritten derivation on grid paper, organized into two main sections separated by a horizontal line. The top section contains several lines of algebraic manipulation, including fractions and square roots, with some terms circled in red and blue. The bottom section starts with a definition of a function $f(x)$ and then proceeds to show a series of steps, including a substitution $x = \frac{1}{y}$ and a final result that appears to be $f(x) = \frac{1}{x}$. The handwriting is in black ink, with some red and blue ink used for emphasis or correction.

14. Proposition 3: paper $\rightarrow$ Lean $\rightarrow$ proof

Paper statement. If $0 \leq p_k^1 \leq p_k^2 \leq 1$, then $\Delta C(s) = \sum_{k \in U} [(1 - p_k^1)^{s+1} - (1 - p_k^2)^{s+1}] \geq 0$; when $U \neq \emptyset$ and $p_k^1 = 0 < p_k^2 < 1$, the gap is strictly increasing and concave.

Lean translation and proof fragment.

```
theorem dynamic_cumulative_language_effect :
  dynamic_cumulative_language_effectSpec := by
  intro Language _ unfamiliar p1 p2 hHazards
  dsimp; constructor
  •intro s -- weak gap:  $\Delta C(s) \geq 0$ 
    apply Finset.sum_nonneg; intro k hk
    have hkB := hHazards k hk
    have h0 :  $0 \leq 1 - p_2 k$  := by linarith
    have hle :  $1 - p_2 k \leq 1 - p_1 k$  := by linarith
    have := pow_le_pow_left0 h0 hle (s + 1)
    linarith
  •rintro ⟨hNe, hClosed⟩ -- closed frontier  $p_1=0 < p_2$ :
    -- strictMono_nat_of_lt_succ + pow_lt_pow_right_of_lt_one0 => strict growth + concavity
```

Plain English. A higher first-use hazard lowers every survival probability $(1 - p)^{s+1}$; summing those nonnegative differences gives $\Delta C(s) \geq 0$. Lean checks this *exactly* under the paper's stated hypothesis $p^2 \geq p^1$; the strict-increase and concavity half is proved on the closed-frontier benchmark $p^1 = 0 < p^2$, precisely the domain the proposition conditions on—no extra assumption is smuggled in.

Verified build

lake build QX26AgenticDelegation: PASS (8318 jobs). Five of five selected endpoints compile with no sorry; EconCSLib -fast check also passed. Independent source-fidelity review is still pending, so status is partial.

15. Verdict

The model cleanly shows how delegated execution can expand a **production** frontier through a lower entry threshold.

The data show the predicted activation spike, accumulation pattern, and specialist headroom signature. They do not causally separate delegation from project-driven adoption, and they do not show skill acquisition.

Paper, derivations, and sources: <https://github.com/johnbarraza/ai-03-quispe>